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COMPUTTER ARCHITACTURE (ADVAN)WK2

1.Scenario Analysis:

Instruction-Level Parallelism (ILP) is a technique used by modern computer processors to execute multiple instructions at the same time, improving performance and reducing the time needed to complete programs. A good example of ILP is **superscalar execution**, where a processor has multiple execution units that allow it to process several instructions during one clock cycle. For example, while one instruction is performing an addition, another instruction can load data from memory, and a third can perform a comparison. The processor checks whether these instructions are independent and can safely be executed simultaneously. If the instructions do not depend on each other, they are issued to different execution units. This increases the number of instructions completed per second and makes better use of the processor's resources. Superscalar execution is commonly found in modern CPUs used in computers, smartphones, and other electronic devices.

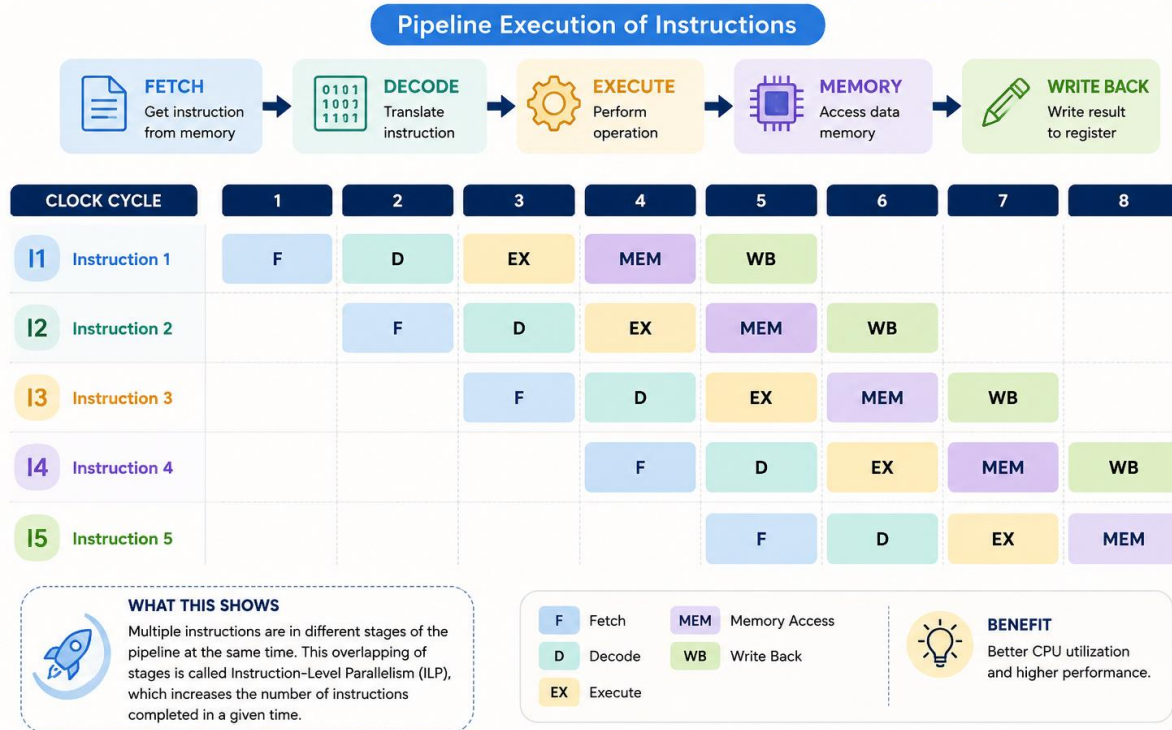
2. Concept Research: Branch Prediction

Branch prediction is an important technique used in Instruction-Level Parallelism (ILP) to improve processor performance. It allows the CPU to predict which direction a conditional branch, such as an **if-else statement**, will take before the condition is fully evaluated. This helps the processor continue fetching and executing instructions without waiting. If the prediction is correct, the processor saves time and keeps its pipeline busy. If the prediction is wrong, the incorrectly executed instructions are discarded and the correct instructions are loaded, causing a small performance penalty. Modern processors use advanced branch prediction methods based on previous program behavior. Overall, branch prediction increases instruction throughput and makes CPUs faster and more efficient.

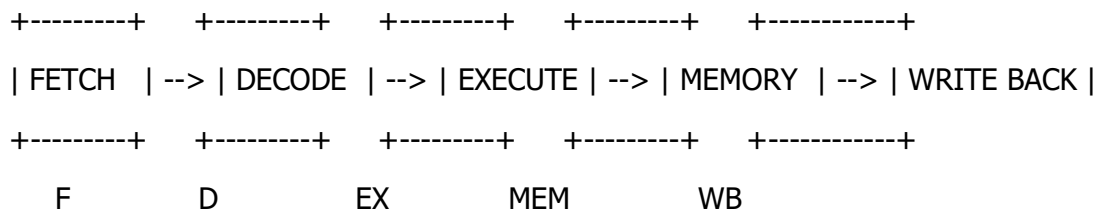
3. Tool Practice: ILP Pipeline Diagram

For this tool practice, I created an ILP pipeline diagram showing how instructions can be processed through different stages of a processor. The diagram includes stages such as **Fetch, Decode, Execute, Memory, and Write Back**. It demonstrates how multiple instructions can overlap during execution instead of waiting for one instruction to finish completely. This activity helped me understand how instruction-level parallelism improves CPU performance by keeping different processor stages active at the same time. I learned that pipelining can increase the number of instructions completed within a given period. Overall, creating the diagram made the concept of ILP easier to understand and showed how modern processors achieve faster execution.

INSTRUCTION-LEVEL PARALLELISM (ILP)



4. ILP Diagram



Instruction-Level Parallelism (ILP) allows a processor to execute multiple instructions simultaneously by overlapping different stages of execution. The pipeline consists of five main stages: Fetch, Decode, Execute, Memory, and Write Back. During the Fetch stage, the processor retrieves an instruction from memory. In Decode, the instruction is interpreted. Execute performs the required operation, while Memory accesses data if needed. Finally, Write Back stores the result in a register. Because different instructions can occupy different stages at the same time, the processor works more efficiently. This improves CPU performance, increases throughput, and reduces the time required to complete tasks.